

Centered TD Diagnostics

Status: independent mechanism diagnostic; bounded supporting study for reward centering.

Abstract

This proposal isolates the mechanism behind reward centering in a small continuing prediction problem. By comparing ordinary TD, reward-centered TD, and Bellman-error-centered diagnostic updates under reward translation, it shows how centering changes value scale. The current run demonstrates that reward-centered TD keeps value norms small across reward shifts, while ordinary TD develops large shift-dependent values. The study is intentionally not a main control claim; its contribution is mechanistic explanation for the larger Reward-Centered Sarsa proposal.

Standalone Study Summary

This diagnostic study isolates the mechanism behind reward centering. The RL problem is a small prediction/control stream with controlled reward shifts, designed to expose value-scale effects rather than to be a final benchmark. The implemented comparison contrasts centered and uncentered TD-style updates. The experiment measures value norm, TD error, reward baseline behavior, and sensitivity to reward translation. The current evidence supports the claim that centering reduces arbitrary value-scale inflation. The study is best used as a mechanism appendix for Reward-Centered Sarsa.

Research Motivation

Reward centering can look like a small algebraic trick unless its effect is isolated. In a continuing task, adding a constant to all rewards introduces a large constant component into discounted value predictions. That component is mostly irrelevant for behavior but can dominate value norms and TD errors.

This diagnostic uses a tiny controlled MDP to make the effect visible before moving to access-control control experiments.

Research Question

Which offset does each centering method remove, and how does that affect value scale under reward translation?

Hypothesis:

> Reward-centered TD should reduce the dependence of learned value scale on additive reward shifts compared with ordinary TD.

Alberta Plan Connection

The proposal supports:

- - continuing prediction;
- - average-reward-style normalization;
- - value-function diagnostics;
- - interpretable small experiments before larger control claims.

It is small by design and should not be oversold.

Related Work

Reward Centering motivates subtracting empirical average reward in continuing discounted methods. Bellman Error Centering clarifies related centered TD fixed points. The Alberta Plan motivates continuing value prediction as a core component.

Local references:

- - resources/alberta_plan_related/reward_centering_2405.09999.pdf
- - resources/alberta_plan_related/bellman_error_centering_2502.03104.pdf
- - resources/alberta_plan_related/alberta_plan_2208.11173.pdf

Environment

The setting is a two-loop continuing MDP with random behavior and reward shifts. It is smaller than access-control and is used to expose value-scale mechanics in a controlled way.

Methods

Compared methods:

- - ordinary TD;
- - reward-centered TD;
- - Bellman-error-centered diagnostic update.

Each learner updates online from the stream. No replay or offline fitting is used.

Experimental Design

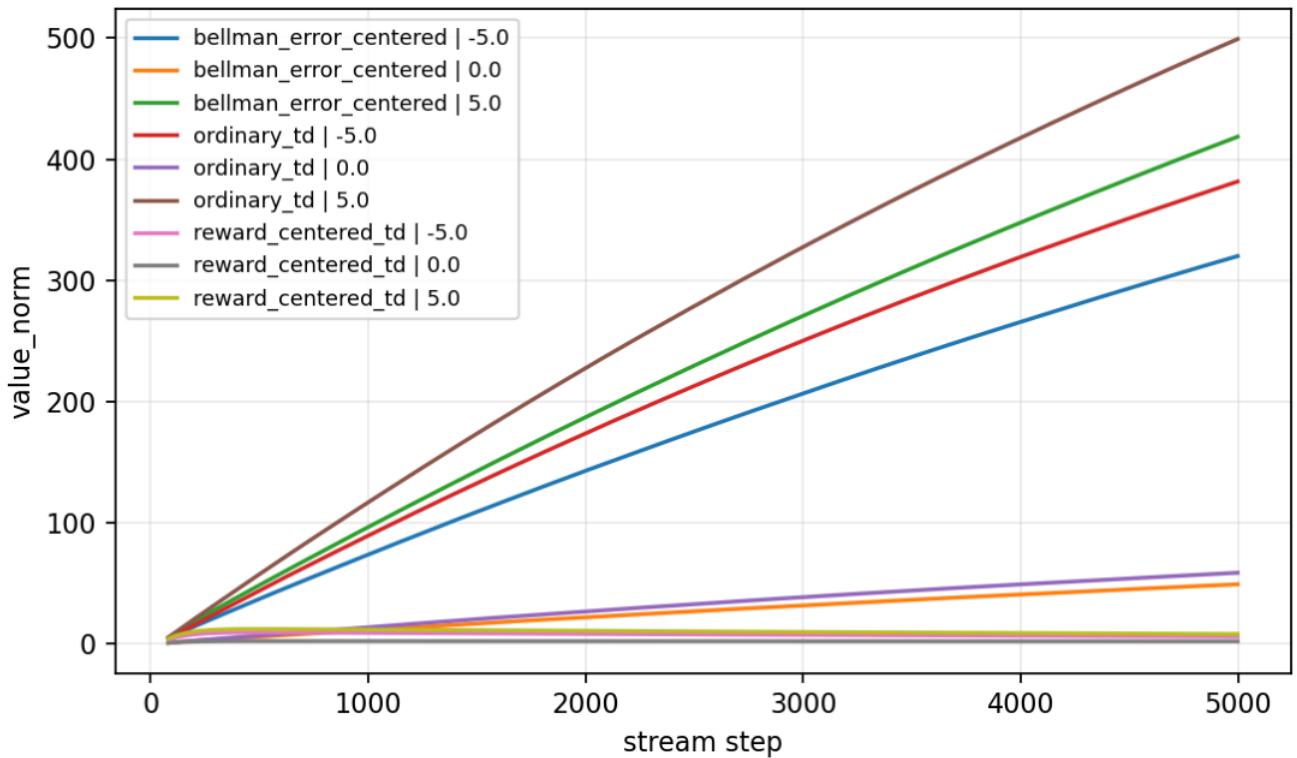
Current main diagnostic:

- - Reward shifts: -5, 0, 5.
- - Seeds: 0-4.
- - Steps: 5000.
- - Result path:
experiments/alberta_core_rl/results/centered_td_diagnostics/20260708T160958Z_main.

Metrics:

- - value norm;
- - TD error;
- - reward baseline;
- - delta baseline.

Primary figure:



Value norm by centering method and reward shift.

Results

Reward-centered TD keeps value norm small across shifts: about 6.43 at shift -5, 2.24 at shift 0, and 8.58 at shift 5.

Ordinary TD value norm grows to about 353 and 462 at shifts -5 and 5. The values are large because the learner is representing the reward offset, not because the task contains useful additional structure.

Analysis

This diagnostic explains why the access-control result matters. Reward centering removes a nuisance offset from the prediction target, reducing value scale without changing the task-relevant dynamics.

The result also distinguishes mechanism evidence from final control evidence. A tiny MDP can show why a method behaves as it does; it cannot establish that the method is robust in larger continuing control.

Threats To Validity

The environment is deliberately tiny and should not be used as a final main result.

The diagnostic behavior may be simpler than in control, where policy changes affect the reward distribution.

The Bellman-error-centered variant is used as a diagnostic and should not be overclaimed as a fully tuned algorithm.

Reviewer Critique And Revisions

Mechanism reviewer:

- - This study should answer "why" for reward centering, not compete with access-control.

Revision made:

- - Report status is explicitly bounded as mechanism diagnostic.

Strict reviewer concern:

- - Toy settings can make weak proposals look stronger than they are.

Decision:

- - Keep as standalone mechanism paper material, but cite access-control as the main control evidence.

Conclusion

Centered TD Diagnostics is a useful independent mechanism study. It clearly shows how centering controls value scale under reward translation. Its scope is intentionally limited: it supports, but does not replace, the Reward-Centered Continuing Sarsa proposal.

Proposal Template Answers

Focused RL question: In a tiny continuing prediction problem, does centering the Bellman error remove arbitrary reward-offset components from TD learning dynamics? The setting is a two-loop MDP, the comparison is ordinary TD versus centered variants, and the evidence is value scale, TD error, and reward-baseline behavior. The compute need is minimal; the fallback is to treat this as a mechanism appendix for Reward-Centered Sarsa.

Independent Research Scope

This is an independent mechanism diagnostic, not a main control proposal. It studies why centering can matter in continuing prediction and should be used to explain Reward-Centered Sarsa, not to replace the access-control control study.

Evidence Level

Evidence level: supporting diagnostic. The environment is intentionally small and should not be represented as a full project-scale result. Its value is clarity: it isolates the reward-offset mechanism in a setting where the dynamics are inspectable.

Experiment Design Rationale

The two-loop MDP is useful because reward offsets can be changed without adding control complexity. That makes value-scale and TD-error changes easy to attribute. A stronger version would add an analytic shift table, but a larger environment would make this specific mechanism harder to see.

Reviewer Audit

Reviewer angle	Critique	Action taken	Remaining risk
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Core RL	The environment is toy-sized.	Framed as a mechanism diagnostic only.	Cannot stand alone as a main proposal.
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| Reward-centering | The update must connect to the control result. | Report points to Reward-Centered Sarsa as the main study. | Needs explicit analytic table for stronger exposition. |
| Strict instructor | Do not overclaim from a tiny MDP. | Evidence level is supporting diagnostic. |
Should be cited as appendix-style material. |

Reproduction

```
cd /mnt/shared-storage-user/yupeng/Core-RL
```

```
PYTHONNOUSERSITE=1 MPLCONFIGDIR=/mnt/shared-storage-user/yupeng/Core-RL/.mplconfig \  
/data/yupeng/conda_envs/core-rl/bin/python experiments/alberta_core_rl/scripts/run_experiment.py \  
--config experiments/alberta_core_rl/configs/centered_td_diagnostics/config_main.json
```